

Aaditya Jaiswal

CH.SC.U4CSE25276

Linked List

Q.No:1

Question description

Professor Shiva decided to conduct an industrial visit for final year students, but he set a condition that if students received a passing grade in the surprise test, they would be eligible to go on the industrial visit.

He asked the students to study a topic linked list for 10 minutes before deciding to conduct a surprise test.

Professor-mandated questions, such as the deletion of nodes with a certain data D, are now being asked.

For example

if the given Linked List is 5->10->15->10->25 and delete after 10 then the Linked List becomes 5->15->25.

Constraints

1 < N < 100
1 < D < 1000

Input Format

First line contains the number of data-N.
Second line contains N integers(the given linked list).
Next line indicates the node data D that has to be deleted.

Output Format

Single line represents the linked list after required elements deleted.

Logical Test Cases

Test Case 1

INPUT (STDIN)

6
7 9 8 8 4 2
8

EXPECTED OUTPUT

Linked List:->7->9->4->2

Test Case 2

INPUT (STDIN)

10
5 3 7 2 7 9 8 8 4 2
2

EXPECTED OUTPUT

Linked List:->5->3->7->7->9->8->8->4

Mandatory Test Cases

Test Case 1

KEYWORD

struct node

Test Case 2

KEYWORD

node *next;

Test Case 3

KEYWORD

void create()

Test Case 4

KEYWORD

p2=p2->next;

Test Case 5

KEYWORD

void del()

Code:

```
#include <stdio.h>
#include <stdlib.h>

struct node
{
int data;
```

```

struct node *next;
};

struct node *head = NULL;
struct node *tail = NULL;
void create()
{
int n, data, i;
struct node *newNode;

scanf("%d", &n);

for (i = 0; i < n; i++)
{
scanf("%d", &data);

newNode = (struct node *)malloc(sizeof(struct node));

newNode->data = data;
newNode->next = NULL;

if (head == NULL)
{
head = newNode;
tail = newNode;
}
else
{
tail->next = newNode;
tail = newNode;
}
}

void del()
{
int D;
struct node *temp;
struct node *prev;
struct node *p2;

scanf("%d", &D);
while (head != NULL && head->data == D)
{
temp = head;
head = head->next;
free(temp);
}

if (head == NULL)
return;

prev = head;
p2 = head->next;

```

```

while (p2 != NULL)
{
if (p2->data == D)
{
prev->next = p2->next;
free(p2);
p2 = prev->next;
}
else
{
prev = p2;
}

p2 = p2->next;
}
}
}

void display()
{
struct node *temp = head;

printf("Linked List:");

while (temp != NULL)
{
printf("->%d", temp->data);
temp = temp->next;
}

printf("\n");
}

int main()
{
printf("Input value:\n");
create();
del();
printf("Output:\n");

display();

return 0;
}

```

Output:

Test_case 1:

00/ linked list/ 1

Input value:

6

7 9 8 8 4 2

8

Output:

Linked List:->7->9->4->2

Test_case 2:

Input value:

10

5 3 7 2 7 9 8 8 4 2

2

Output:

Linked List:->5->3->7->7->9->8->8->4